

Inhibitor Tolerance: Data, Representation and Model Plan

What the four chemogenomic datasets can share as input, which molecular encoders carry the compounds, and how one decoder reads four response scales

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Abstract

Five served datasets hold 7.48 million chemogenomic records, and training one model across them requires a single input representation and a single decoder. They disagree on four axes. Ploidy runs from haploid through homozygous diploid to heterozygous diploid. Compound panels run from 41 to 5,170. Dose runs from an unquantified target inhibition through a nine-order-of-magnitude molar range to a single fixed concentration. And the response is a log ratio, a z score or a sensitivity score depending on the source. Every number here is measured from the served records through their loaders.

The genotype axis resolves cleanly. The schema already carries heterozygosity as a genome-wide ploidy baseline and a per-locus copy-number deviation, and a per-gene functional dose, copies present over copies in the reference, is lossless for these datasets because every perturbation is a full deletion or a one-of-two variant. It is learnable: Hoepfner supplies 648,977 gene-by-compound cells measured both heterozygous and homozygous. The gap is entirely in the tensor layer, which reads only a gene name and no environment field at all. The dose axis does not resolve, so dose enters as a basis token and a masked log molar value rather than one number. The physical fields, temperature, aerobicity, duration and pH, are constant inside every dataset, so they need no channel and the dataset token carries them. Chemically, pooling buys the phenolic inhibitors and not the small alcohols, and Wildenhain supplies essentially all of the gain; twelve molecular encoders cover all 5,472 dosed compounds, the 37 medium components and 894 of the 1,378 Yeast9 species from one embedding table.

Two decisions are taken. The Hillenmeyer homozygous arm is dropped, leaving four datasets and 6.39 million records. And the functional dose scales the existing cross-attention perturbation operator through a factor that equals one for a deletion, so every current dataset’s model is unchanged and heterozygosity is the only new behavior. The decoder is one scalar head with a per-source dataset token at the readout, the mechanism experiment 030 smoke-tested. The first round is four arms on a fixed compound-cold evaluation within Vanacloig, scored against the reliability ceiling of every compound.

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1 The question and the decisions ■

Five chemogenomic datasets are served and together hold 7.48 million records. Training one model on them is attractive for an obvious reason and blocked by a less obvious one. The attraction is scale and chemical breadth: no single dataset has both a genome-wide gene panel and a large compound panel. The block is that a model takes one input format and one output format, and the five datasets disagree about what a strain is, what a dose is, and what a response is.

Three axes carry the input disagreement. Ploidy runs from haploid single deletions through homozygous diploids to heterozygous diploids where one wild-type copy survives. Compound panels run from 41 to 5,170. Dose runs from an unquantified target inhibition, through a molar range spanning nine orders of magnitude, to a single fixed concentration applied to every compound. The output disagreement is a fourth axis: the response is a log ratio in two datasets, a z score in two, and a sensitivity score in one, read from a pooled barcode assay in four and from liquid optical density in one.

The application target is Vanacloig-Pedros 2022 (10.1128/spectrum.02017-22), the only served screen that doses the industrial hydrolysate panel at a matched target inhibition and the only one in which isobutanol appears as a stress. The others enter as partners: Hillenmeyer 2008 (10.1126/science.1150021) and Hoepfner 2014 (10.1016/j.micres.2013.11.004) for record count and gene coverage, Wildenhain 2015 (10.1016/j.cels.2015.12.003) for compound diversity.

What this document is. It is the planning document for the modeling phase, in one place: what the earlier data exploration established about ceilings and transfer (Section 4), what a shared input can absorb on each axis (Sections 6 to 8), which molecular encoders carry compounds into the model and how far they reach (Sections 9 and 10), how the channels enter the cell graph transformer and what one decoder over four response scales looks like (Section 11), and the representation itself with its implementation steps (Section 12).

Two decisions taken. The measurements below left two choices open, and both are now settled. The Hillenmeyer homozygous arm is dropped: its 1.09 million records bring one non-exact chemical neighbor above 0.5 to the application panel, and its heterozygous sibling keeps the gene coverage. Four datasets remain, 6.39 million records. And the per-gene functional dose scales the existing cross-attention perturbation operator rather than entering as a separate feature, because the operator is already a soft mechanism and a scale factor that equals one for a deletion leaves every current dataset’s model unchanged.

Scope. This phase handles two perturbation kinds only: the full deletion and the one-of-two copy-number variant, which are the only kinds any of the four datasets contains. Alleles off the dosage axis, temperature-sensitive, decreased-abundance-by-mRNA-perturbation (DAmP) and CRISPR interference, are not represented here and are not needed by these datasets; the representation is written so they can be added without changing what is built now.

2 The datasets ■

Table 1 puts the five served datasets on the axes that matter for a shared input. Record counts and gene counts are read from the flattened served records; compound counts are distinct InChIKeys with a dosed record; the dose columns are the stored basis and the share of records whose unit converts to molar without a molecular weight. The Hillenmeyer homozygous arm is kept in the table because the measurements that led to dropping it are made against it.

Table 1: The five served chemogenomic datasets. Records and genes come from the flattened served records, compounds are distinct InChIKeys with a dosed record, and the last column is the share of records whose dose unit converts to molar. The Hillenmeyer homozygous arm is dropped from the model (Section 1).

dataset	records	genes	compounds	ploidy	dose basis	molar
Vanacloig 2022	143,218	3,598	41	haploid	IC30, fixed	0%
Hillenmeyer HOM	1,088,620	4,675	114	diploid	fixed	84%
Hillenmeyer HET	2,698,797	5,825	290	diploid	fixed	96%
Hoepfner 2014	3,124,319	5,842	148	diploid	IC30	100%
Wildenhain 2015	428,206	242	5,170	haploid	none	100%

Three observations set up the rest of the document. Hoepfner is the largest store and the only one holding both ploidy arms, 1,759,255 heterozygous records and 1,365,064 homozygous, over a shared gene universe. Wildenhain has eighteen times the compound panel of the next largest and a fifth of the gene panel. Vanacloig is the smallest and is the application target, and it is the only dataset whose dose is not a number.

Two corrections worth recording, because both were wrong in earlier readings of the raw files. The Hoepfner score matrices are genes by screens, not screens by genes: the 6,681 rows are the genes. And the roughly 5,900 columns per

file are not experiments run in duplicate. A census of every column header (Table 9) finds 2,905 adjusted-score columns in the HIP file and 2,904 in the HOP file, each carrying a distinct experiment, plus a companion gene-wise z score for every one of them. The doubling is the second normalization, not a second measurement, and Section 5 takes up what that means for Hoepfner’s prediction ceiling.

The physical and medium fields are dataset identity. The served record carries the medium, the temperature, the aerobicity, the duration, the pH when it was set, and the solvent. Table 2 reads every one of them from the records of all five datasets. The result is that none of them varies inside a dataset in a way a model could learn from. Temperature is 30 C wherever it is stated and is stated for 1 to 2 percent of Hillenmeyer records, the few exceptions being its heat and cold arms. Aerobicity separates Vanacloig, which is anaerobic, from everything else. Duration is stored in hours by two datasets and in generations by three, with no dataset varying it except Hillenmeyer’s generation arms. pH is stated only by Vanacloig, at 5.0, and by 2 percent of Hillenmeyer homozygous records. Across the five, no stated pH or duration in hours appears in more than one dataset.

Table 2: The physical and medium environment fields of every served record, by dataset. A cell lists the distinct stated values; where fewer than all records state one, the share that do is given.

field	Vanacloig	HOM	HET	Hoepfner	Wildenhain
medium base	SynBase (SynH3 minus)	SC, SD, YP, YPD	SD, YP, YPD	YPD	SC
temperature, C	30	23, 25, 37 (2% stated)	20, 23, 37 (1% stated)	30	30
aerobicity	anaerobic	aerobic	aerobic	aerobic	aerobic
duration, h	48	not stated	not stated	16 (44% stated)	18
duration, generations	6.5	0, 5, 10, 15, 20	0, 5, 10, 15, 20	5, 20	not stated
pH	5	7.5, 8 (2% stated)	not stated	not stated	not stated
solvent	not stated	not stated	not stated	DMSO	DMSO
measurement	log2 ratio	z score	log2 ratio	sensitivity	z score
assay	barcode pool	barcode pool	barcode pool	barcode pool	liquid OD

The consequence for the representation is direct. A temperature channel, an aerobicity channel or a pH channel would take a constant value inside each dataset and a different constant in each, which makes it the dataset identity written a second way. The representation in Section 12 therefore gives the physical fields no channel of their own and lets the dataset token carry them. The fields stay in the record, and the day a served dataset varies temperature within itself the channel is a one-line addition.

The same table shows the output disagreement. The response is a log ratio in Vanacloig and Hillenmeyer HET, a z score in Hillenmeyer HOM and Wildenhain, and a sensitivity score in Hoepfner; four datasets read it from a pooled barcode competition and Wildenhain from liquid optical density. Section 11 takes this up as the decoder question.

3 What a record is and what is done to it ■

Before any argument about pooling, this section fixes what the data actually looks like and what the model is asked to do with it, because the rest of the document is unreadable without both. Table 3 defines every term that appears in a later figure label or table header.

The task is regression on a continuous number. It is worth stating plainly, because the sign conventions in the next section invite the opposite reading. The model predicts a real-valued growth response for a strain under a compound. Nothing is thresholded into sick and healthy, no class label is formed, and no cutoff appears anywhere in the pipeline. The reason the document talks about which SIGN means a sick strain is that the five sources disagree about that sign, and a pooled target has to resolve the disagreement before it can be regressed on. Resolving it costs nothing: the fix is a multiplication by plus or minus one, which is a bijection on the real line and discards no information. Were the target binarized, the compound-level correlations that every score in this document reports would not be the right way to measure it.

One real record from each dataset, carried through. Table 4 takes an actual served record from each of the five and prints it at every step. Reading a row left to right gives what the strain is, what it was dosed with, what the source reported, and what the pooled model would be asked to predict.

Three things in that table are worth naming, because each is a place the datasets differ in a way a reader would otherwise have to infer.

The Vanacloig strain perturbs four genes, not one. Three of them, PDR1, PDR3 and SNQ2, are the drug-efflux regulators deleted to build the sensitized host, and they are the same three in every Vanacloig record; the fourth is the

Table 3: Controlled vocabulary, alphabetical. Each term is used in a figure label or a table header later in the document.

term	meaning
calculated logP	The octanol-water partition coefficient estimated from structure by the Crippen method in RDKit. It measures lipophilicity: a negative value is water-preferring, a positive value is fat-preferring. Nothing is measured here; it is computed.
ceiling	The highest correlation any model could reach against the noise-free response, given how noisy the measurement is. It is the square root of the reliability. A score of 0.31 against a ceiling of 0.84 means the model captured 37 percent of what the data allows, not 31 percent of the truth.
cell	One (gene, compound) pair. Used when counting what two datasets both measured, so that a comparison is between the same gene under the same compound.
compound cold-start	An evaluation that holds out every record of one compound at a time, so the model must say something about a molecule it never saw dosed. This is the split that matters for predicting a hydrolysate inhibitor.
HIP, HOP	The two assays of a pooled chemogenomic screen. HIP (haploinsufficiency profiling) uses heterozygous diploids, one wild-type copy left. HOP (homozygous profiling) uses homozygous diploids, both copies gone.
HOM, HET	The two arms of Hillenmeyer 2008, its homozygous and heterozygous collections, released as separate files with different readouts and handled here as separate datasets. The homozygous arm is the one dropped.
MADL score	Median absolute deviation logarithmic score, the per-experiment readout Hoepfner reports: the strain’s log abundance ratio minus the median over all strains in the sample, divided by the median absolute deviation of that same set.
orientation factor	Plus or minus one per dataset, multiplied into the response so that afterwards a more negative value means a sicker strain everywhere. It is a sign flip on a continuous number, not a threshold and not a class label.
reliability	The share of a measurement’s variance that is real rather than noise. Estimated either from a served standard error or from the agreement between two independent measurements of the same condition.
sensitized host	A strain whose drug-efflux regulators are deleted so that compounds accumulate instead of being pumped out. Vanacloig deletes PDR1, PDR3 and SNQ2, so every one of its genotypes carries those three deletions plus the gene being queried, which is why its records perturb four genes rather than one.

Table 4: One real served record from each dataset, carried through the two target transformations. The strain column gives the queried gene with its common name, and for Vanacloig also the three sensitized-host deletions every one of its genotypes carries. o_k is the orientation factor and $\tilde{y}$ the standardized target. The records were chosen as the strongest response of a gene measured in all five, so the arithmetic is visible rather than sitting at zero.

dataset	strain	ploidy	compound	reports	raw	o_k	$\tilde{y}$
Vanacloig 2022	YBR058C (UBP14) + host	hap	crystal violet	log2 ratio	-3.40	+1	-4.80
Hillenmeyer HET	YBR058C (YBR058C)	dip	nocodazole	log2 ratio	+1.92	-1	-4.30
Hoepfner 2014	YBR058C (YBR058C)	dip	Epothilon B Deriva	sensitivity score	-7.17	+1	-4.48
Wildenhain 2015	YMR263W (SAP30)	hap	3-octadecanamido-2	z score	-81.84	+1	-10.87
Hillenmeyer HOM	YBR058C (YBR058C)	dip	mitomycin c	z score	+56.29	-1	-15.95

gene being queried. That is what the four-gene perturbation count in Table 1 means, and it is the reason a Vanacloig genotype contributes four indices to the perturbed set where the others contribute one.

One gene appears in four of the five rows on four different scales. UBP14 under crystal violet reads -3.40 as a log ratio, the same gene under nocodazole reads $+1.92$ as a log ratio pointing the other way, under an epothilone derivative -7.17 as a sensitivity score, and under mitomycin C $+56.29$ as a z score. These are not phenotypes of different magnitude; they are the same kind of measurement reported on scales that differ by a factor of twenty and, for two of the five, with the opposite sign. The fifth row is a different gene because Wildenhain’s panel is 242 genes and does not contain this one, which is the narrowness Table 1 records.

After both steps all five land in the same range and the same direction. Every standardized value in the table is negative, which under the shared convention means every one of these records is a strain that grew badly, which is what picking the strongest response per dataset was meant to produce.

The two steps, and why in that order. For a record from dataset k with served response y ,

$$\tilde{y} = o_k \cdot \frac{y - \text{med}_k}{\text{sd}_k},$$

where $o_k \in \{-1, +1\}$ is the orientation factor and med_k and sd_k are the median and standard deviation of dataset k on the TRAINING split only, so no held-out value informs the scale. Orientation comes first because the median and the standard deviation are computed on the oriented values; the standard deviation is unaffected by the sign, but stating the order keeps the two from being applied inconsistently. Three of the five sources already store a sick strain as negative and take $o_k = +1$, and only the two Hillenmeyer arms flip.

Section 5 is where o_k comes from, and it is not a preference. The five sources declare opposite conventions in their own words, and ignoring that puts two records of the same gene under the same compound on opposite sides of zero.

4 What the data exploration established ■

The earlier document, `notes-tex/031-inhibitor-tolerance-data`, compared Vanacloig with the two Hillenmeyer arms axis by axis. Three of its results bound everything planned here, and the linear and nearest-neighbor baselines run since then say how much of the bound a simple model reaches.

Replication sets the ceiling, and the served error is honest for one dataset only. The reliability index of a condition is one minus the mean squared served standard error over the variance of the response, and its square root is the ceiling on the correlation any model can reach with the noise-free response. Table 5 places that index beside the raw replicate agreement of the same conditions and beside what the Spearman-Brown lift of the single-replicate reliability predicts for the mean of k replicates. For Vanacloig the index sits 0.04 above the prediction and its median ceiling is 0.84, with three of 41 conditions carrying no signal at all, 5-hydroxymethylfurfural among them. For both Hillenmeyer arms the index sits 0.28 and 0.36 above the prediction, so their served error understates the noise and the raw replicate agreement, 0.39 and 0.26, is the number to plan against. Hoepfner and Wildenhain serve no uncertainty on most records, so no ceiling exists for them yet.

Table 5: Reliability per dataset, medians over conditions. Replicate rho is the raw between-replicate Spearman over genes; rel. single is the single-replicate reliability it implies; k the replicate count; predicted is the Spearman-Brown reliability of the mean of k ; index is the served-error reliability; ceiling is $\sqrt{\text{index}}$; no signal counts conditions whose index is at or below zero.

dataset	conditions	replicate rho	rel. single	k	predicted	index	index minus predicted	ceiling	no signal
Vanacloig 2022	41	0.33	0.39	3	0.66	0.70	+0.04	0.84	3
Hillenmeyer HOM	140	0.39	0.42	4	0.59	0.92	+0.28	0.96	0
Hillenmeyer HET	308	0.26	0.25	4	0.42	0.84	+0.36	0.92	2

Cross-dataset transfer is a gene prior, not compound-matched transfer. Between Vanacloig and the Hillenmeyer homozygous arm the per-gene mean response correlates at 0.118 and the gene-gene similarity structure agrees eleven null standard deviations from zero, while the median Spearman between matched condition profiles is 0.010. The attenuation limit set by the two reliabilities is 0.55, so the failed transfer is not a noise artifact. Exactly one compound, methyl methanesulfonate, has a per-gene profile that transfers; benomyl, the acetate salt against the free acid, and ferulic acid at two doses do not. Chemical similarity correlates with response similarity across the two datasets at Spearman 0.09 at best (Section 9).

Within Vanacloig, molecule features carry real signal and gene embeddings do not. Table 6 gives the best ridge and nearest-neighbor model per split and target beside the nulls. On the compound cold-start split, leaving one

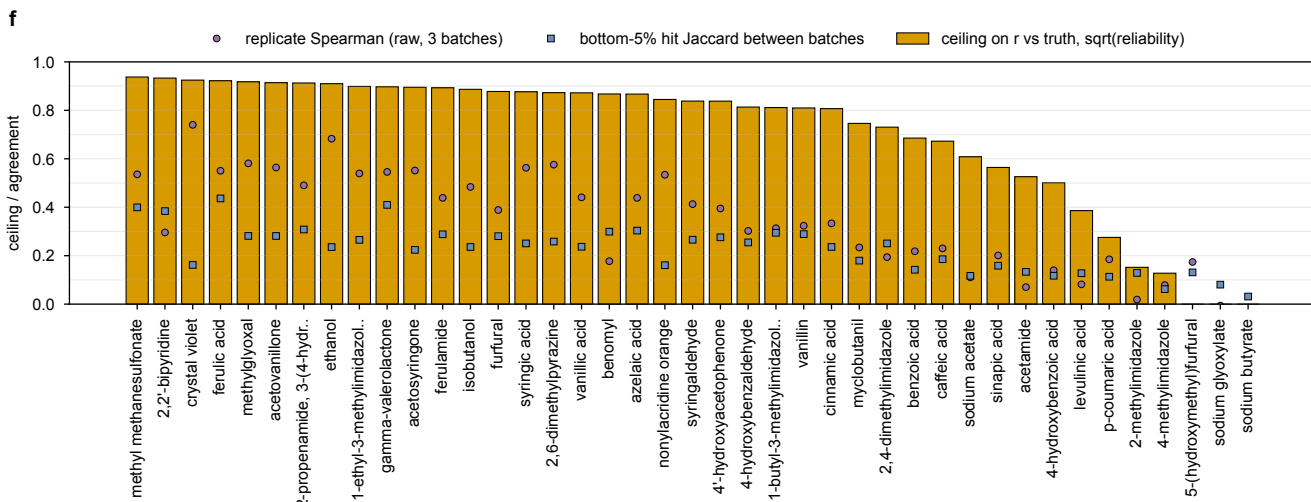


Figure 1: Every Vanacloig compound, ordered by its ceiling on the correlation with the noise-free response (bars, $\sqrt{\text{rel}}$), with the raw three-batch replicate Spearman (circles) and the between-batch hit Jaccard (squares). A compound with no bar has an index at or below zero.

of the 41 compounds out in turn, the centered target subtracts each gene’s mean over the training compounds so that only compound-specific signal remains. There a ridge map from the FCFP4 fingerprint reaches Spearman 0.311 against 0.020 for a random training neighbor, positive in 93 percent of folds, which is 42 percent of the 0.835 ceiling. On the raw target the gene mean alone reaches 0.212 and the best ridge 0.341. On the gene cold-start split, five gene folds by 41 compounds, protein-language-model embeddings reach 0.044, real but 8 percent of the ceiling. Ridge beats every nearest-neighbor variant, and the encoder ranking does not favor the large pretrained models: a plain feature-class fingerprint is best on the centered target.

Table 6: Linear and nearest-neighbor baselines on Vanacloig against the reliability ceiling, medians over folds. For each split and target the best feature set per model class is shown. Compound cold holds out one compound per fold; gene cold holds out a fifth of the genes per fold and scores each compound column. The centered target removes each gene’s training mean.

split	target	model	features	folds	Spearman	Pearson	ceiling	of ceiling
compound cold	centered	ridge	fcfp4 count	41	0.311	0.350	0.835	0.42
compound cold	centered	knn5	unimol v1	41	0.284	0.328	0.835	0.39
compound cold	centered	knn3	fcfp4 count	41	0.283	0.327	0.835	0.39
compound cold	centered	knn1	molformer xl	41	0.280	0.279	0.835	0.33
compound cold	centered	mean train profile	none	41	-0.001	-0.011	0.835	-0.01
compound cold	raw	ridge	mole static	41	0.341	0.393	0.838	0.47
compound cold	raw	knn5	chemberta2 mlm	41	0.312	0.401	0.838	0.48
compound cold	raw	knn3	roberta zinc 480m	41	0.287	0.367	0.838	0.44
compound cold	raw	knn1	fcfp4 count	41	0.281	0.338	0.838	0.40
compound cold	raw	mean train profile	none	41	0.212	0.255	0.838	0.30
compound cold	raw	gene mean	none	41	0.212	0.255	0.838	0.30
gene cold	centered	ridge	prott5	205	0.044	0.068	0.841	0.08
gene cold	raw	ridge	prott5	205	0.044	0.068	0.841	0.08

Two consequences carry into the plan. More than half the predictable variance on the compound-cold target is unclaimed by a linear map, so there is headroom for a model that uses the gene graph. And within-dataset chemistry works far better than cross-dataset chemistry, 0.31 against at most 0.09, which is what one medium, one dose basis and one readout buy; the dataset token and the dose basis token in Section 12 are how the pooled model is told which regime a record came from.

5 Can these datasets be joined at all ■

Pooling is an assertion about the LABEL, not only about the input. Two records that measure the same gene under the same compound have to mean the same thing before one loss can score them both. This section tests that assertion on the served records rather than assuming it, and the test changes what the pooled target has to look like.

A cell here is one (gene, compound) pair. A record enters only if it doses exactly one compound carrying an InChIKey and perturbs exactly one gene, so nothing is averaged across combinations; Vanacloig’s three host deletions are stripped first, leaving the queried gene.

Table 7: What each dataset measures and reports, and the sign its source declares for a sick strain. The standard deviation and skew are of the served response over its cells. The last column is the factor the orientation step applies.

dataset	genes x compounds	cells	dose basis	reports	sd	skew	sick is	o_k
Vanacloig 2022	3,598 x 41	143,218	IC30, fixed	log2 ratio	0.70	-2.6	negative	+1
Hillenmeyer HOM	4,655 x 114	483,279	fixed	z score	2.35	+5.6	positive	-1
Hillenmeyer HET	5,810 x 290	1,651,248	fixed	log2 ratio	0.38	+1.2	positive	-1
Hoepfner 2014	5,839 x 148	861,345	IC30	sensitivity score	1.29	-4.4	negative	+1
Wildenhain 2015	242 x 5,170	428,189	none	z score	7.50	-4.6	negative	+1

The scales differ by a factor of twenty. The served standard deviations run from 0.38 for Hillenmeyer HET to 7.50 for Wildenhain (Table 7, Figure 2a and b). A pooled squared-error loss weights a record by the square of its residual in native units, so without a per-source scale the objective is dominated by Wildenhain and by the tail of Hillenmeyer HOM, whatever the record counts are. Standardizing each source by its own training-split statistics is therefore not a cosmetic step, and the statistics come from the training split alone so that no held-out value informs the scale.

The polarity is not shared, and this is the finding that matters. A more negative number does not mean a sicker strain in every dataset. Each loader records its source’s own definition verbatim, and those definitions split the five into two groups. Vanacloig stores $\log_2(\text{inhibitor/control})$, so depletion is negative. Hoepfner stores an adjusted MADL sensitivity score whose definition reads “negative = hypersensitive, positive = resistant”. Wildenhain stores a growth-inhibition z score where negative is inhibition. Both Hillenmeyer arms store a fitness-defect statistic whose definition reads “positive = fitness defect”.

Two independent checks confirm the split rather than leaving it as a reading of the documentation. First, the heavy tail of each distribution falls on the side its source names: the three negative-is-sick datasets have skew -2.6 , -4.4 and -4.6 , and the two positive-is-sick arms have skew $+5.6$ and $+1.2$ (Figure 2c). Second, the declared polarities predict the SIGN of every pairwise correlation between datasets, and they get all ten right (Table 8). Under a null in which the declared polarity says nothing about the data, ten correct signs out of ten has probability 0.002.

The consequence is concrete and large. As served, Hillenmeyer HOM and Hoepfner correlate at -0.198 over 100,276 shared cells, and Hillenmeyer HET and Hoepfner at -0.118 over 170,837. Pooling without orienting trains a model on contradictory labels at that scale, and the contradiction is worst exactly where the overlap is largest.

Table 8: Every dataset pair: the genes, compounds and (gene, compound) cells both measure, the Spearman correlation on those cells as served, the sign predicted from the two declared polarities, and the correlation after orienting. The predicted sign is correct in all ten pairs.

pair	genes	compounds	cells	ρ as served	sign predicted	ρ oriented
Hillenmeyer HOM / Hillenmeyer HET	4,625	99	407,072	+0.117	+1	0.117
Hillenmeyer HET / Hoepfner 2014	5,802	30	170,837	-0.118	-1	0.118
Hillenmeyer HOM / Hoepfner 2014	4,651	23	100,276	-0.198	-1	0.198
Hillenmeyer HET / Wildenhain 2015	238	123	17,218	-0.018	-1	0.018
Vanacloig 2022 / Hoepfner 2014	3,592	2	6,977	+0.013	+1	0.013
Vanacloig 2022 / Hillenmeyer HOM	3,544	2	6,854	-0.075	-1	0.075
Vanacloig 2022 / Hillenmeyer HET	3,571	2	6,852	-0.023	-1	0.023
Hoepfner 2014 / Wildenhain 2015	237	34	4,787	+0.061	+1	0.061
Hillenmeyer HOM / Wildenhain 2015	199	30	3,292	-0.001	-1	0.001
Vanacloig 2022 / Wildenhain 2015	153	10	761	+0.078	+1	0.078

Agreement after orienting is real, positive, and small. Once oriented, every pair correlates positively, but weakly: the median over the ten pairs is 0.068 and the largest is 0.198, for Hillenmeyer HOM against Hoepfner over 100,276 cells (Table 8, Figure 2e). The two Hillenmeyer arms, which share a laboratory, a medium and a readout family, reach 0.117 over 407,072 cells. Against the application target the numbers are smaller still, because Vanacloig shares only two compounds with each of the Hillenmeyer arms and with Hoepfner, and ten with Wildenhain.

That is the honest shape of the case for joining. These datasets agree about which genes are generally fragile far more than about how a particular gene answers a particular compound. So a shared SCALE is justified and a shared RESPONSE is not: pooling should be expected to buy a gene prior and a wider compound space, not more labels for the same task. Section 11 turns that into a decoder that gives each source its own token, and Section 4 already measured the alternative, which is that within one dataset, one medium and one dose basis, the same molecule features reach 0.31 where they reach at most 0.09 across datasets.

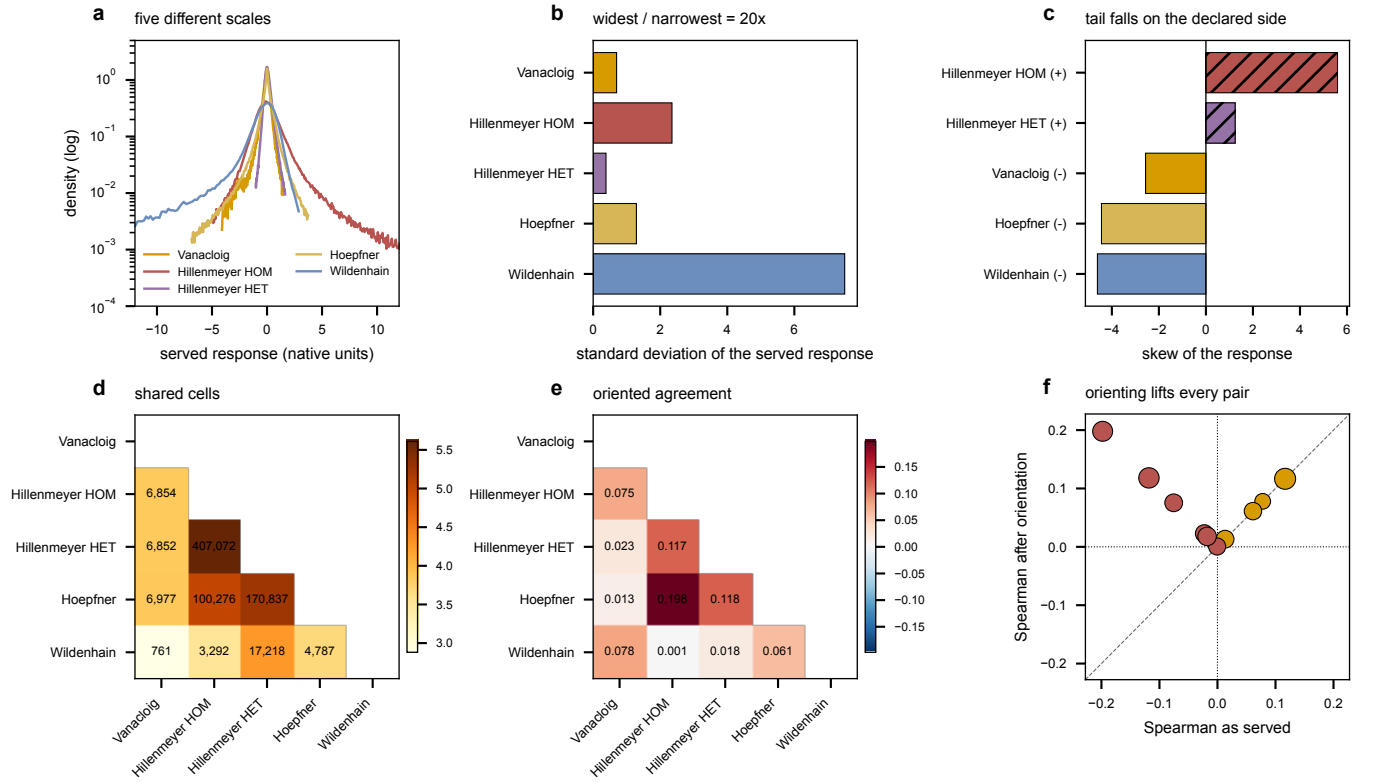


Figure 2: Whether the five datasets can share one target. **(a)** Response density per dataset on a log scale, which is where the phenotype lives. **(b)** Standard deviation of the served response. **(c)** Skew beside the sign each source declares for a sick strain; hatching marks the two datasets whose sources use the opposite convention. **(d)** Shared (gene, compound) cells per pair, log color scale. **(e)** Spearman on those shared cells after orienting. **(f)** The same pairs as served against oriented, point area by shared cells; red points are pairs whose sources disagree about the sign.

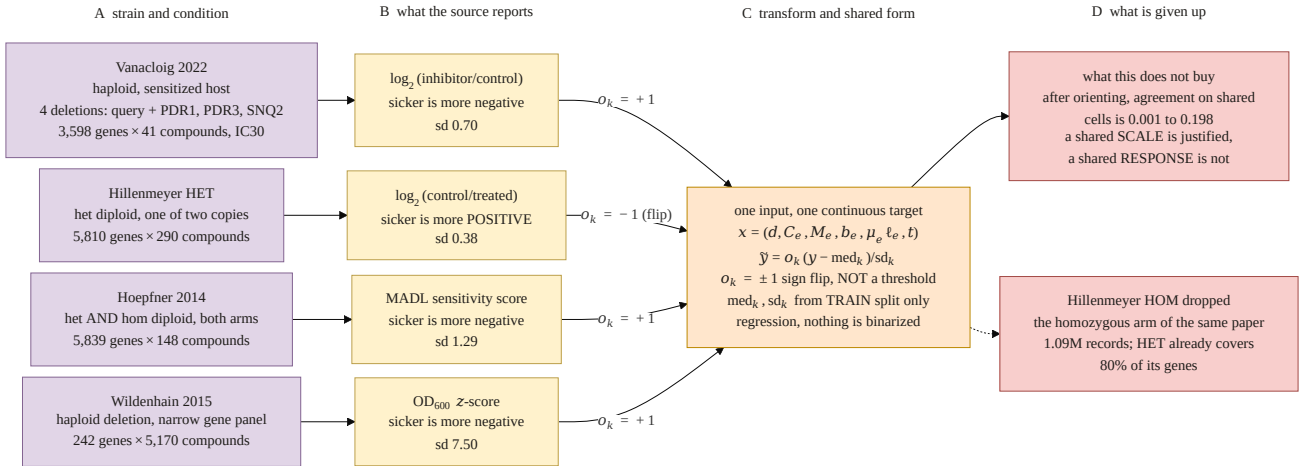


Figure 3: One row per kept dataset, read left to right through four layers. **Layer A**, purple, is what the record IS: the strain, its ploidy, and the size of the gene and compound panel. Vanacloig’s four deletions are the queried gene plus the three efflux regulators that make its host drug-sensitive. **Layer B**, yellow, is what the SOURCE REPORTS, in its own units, with its own sign convention and its own spread; the two $\log_2$ ratios point in opposite directions because one is taken treated over control and the other control over treated. **Layer C**, orange, is the TRANSFORM and the shared form it produces. The orientation factor o_k is a multiplication by plus or minus one, so three datasets pass through unchanged and only Hillenmeyer HET flips; it is a sign flip on a continuous number and never a threshold, and the target stays a real number that the model regresses on. The median and standard deviation come from the training split alone. **Layer D**, red, is what is given up: the homozygous arm of Hillenmeyer 2008, dropped because its heterozygous sibling already covers 80 percent of its genes, and the measured limit that orienting buys a shared scale rather than a shared response. MADL is the median absolute deviation logarithmic score, Hoepfner’s per-experiment readout. Per-dataset counts are in Table 7.

Where the datasets overlap on every other axis. Figure 4 puts the same question to the genotype, chemistry and dose axes. Genes overlap heavily and compounds barely: 98 to 100 percent of Vanacloig’s genes appear in each of the three genome-wide partners, while at most 24 percent of its compounds appear in any one of them. Wildenhain is the exception in both directions, covering 4 percent of the other gene universes and contributing by far the most chemistry. The chemical panels show why the compound gain is uneven: the partner libraries are drug-like at 20 to 40 heavy atoms, and the hydrolysate panel sits at 3 to 15, in a region where a substructure fingerprint is nearly empty. The dose panel repeats Section 7 visually, with Vanacloig carrying no convertible concentration at all.

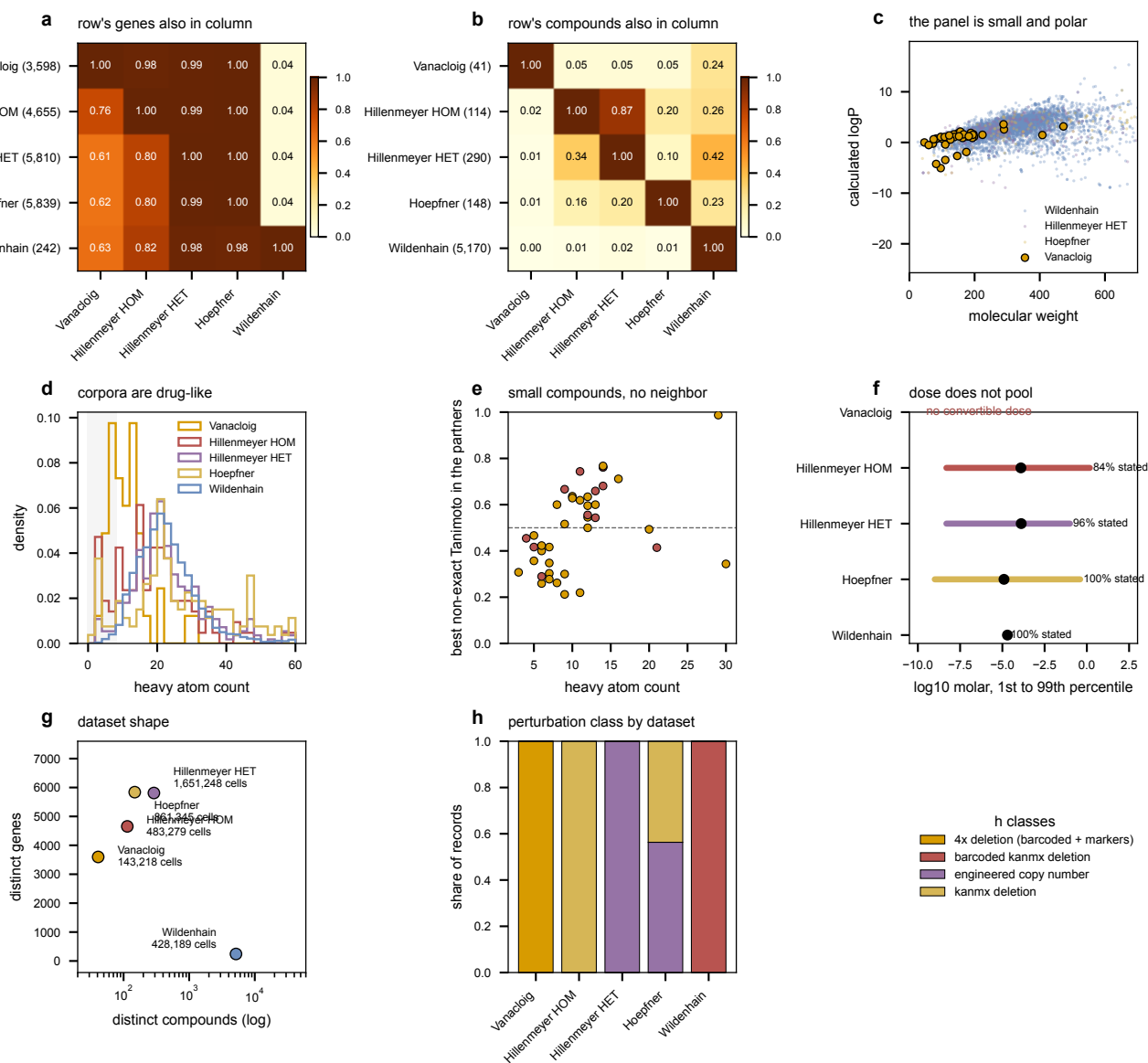


Figure 4: Overlap on every axis a pooled model sees. **(a)** Share of the row dataset’s genes that the column dataset also measures, with each dataset’s gene count. **(b)** The same for compounds. **(c)** Molecular weight against calculated logP for every dosed compound, the Vanacloig panel outlined. **(d)** Heavy atom count; the shaded band below 8 heavy atoms is where a substructure fingerprint carries almost no bits, which is where most of the hydrolysate panel sits. **(e)** Each Vanacloig compound’s best non-exact Tanimoto in the partner union against its size. **(f)** The molar range each dataset spans, 1st to 99th percentile, with the share of records stating a convertible dose. **(g)** Distinct compounds against distinct genes, with each dataset’s measured cell count in its label. **(h)** Perturbation class by dataset.

Per-dataset baselines, because a pooled claim needs a per-dataset reference. Table 10 runs the compound cold-start baselines of Section 4 on every dataset separately, so that any later pooled result has something to be compared against. Leave-one-compound-out is used where the panel allows it and ten folds grouped by compound where it does not, and the scheme is recorded per row rather than left implicit; the two are not interchangeable and a row scored under one should not be read against a row scored under the other as though they were.

Three results matter for the plan. Molecule features carry compound-specific signal in every dataset, not only in the one measured before: each beats its own no-feature null on the centered target. The best score is 0.394 for Wildenhain, 0.311 for Vanacloig, 0.189 for Hoepfner and 0.13 to 0.14 for the two Hillenmeyer arms, though Wildenhain and Vanacloig are not scored under the same scheme and its null is also by far the highest, 0.168 against 0.003 to 0.030 elsewhere,

which is what a panel of 5,170 compounds does to a random neighbor. And which simple model wins is not constant: ridge is better on Vanacloig and Hoepfner, while similarity-weighted nearest neighbors is better on Wildenhain and both Hillenmeyer arms, by the largest margin on Wildenhain where it reaches 0.394 against ridge at 0.249. The earlier finding that ridge beat every nearest-neighbor variant was measured on Vanacloig alone and does not generalize.

One limit on the ceilings in that table. The Hillenmeyer and Wildenhain ceilings rest on the minority of cells that carry a served standard error, 33, 17 and 1 percent respectively, which is the subset measured more than once and therefore not a random sample of the dataset. Only the Vanacloig ceiling, at 97 percent of cells, is estimated from substantially the whole matrix. Hoepfner serves no uncertainty at all, and the next subsection recovers a ceiling for it by a different route.

Hoepfner has no served error, and a ceiling is still recoverable. The absence is real and typed: the loader records it as a provenance gap because the source folds the replicate t -test into the adjusted score rather than releasing a per-cell error. Two different questions follow, and they have different answers (Table 9).

Are the two technical replicates recoverable? No. Each compound was profiled at two wells in the same plate, but a census of every column header in both deposited matrices finds one adjusted-score column per experiment and no experiment id appearing twice. The two wells are already combined inside that column, and the only other column per experiment is the companion z score, which is a second normalization of the same numbers rather than a second measurement.

Is there any repeated measurement at all? Yes, and it is the more useful kind. Fourteen compounds were screened at the same concentration in more than one study, and a study is a separate screen with its own control set, giving 71 condition pairs across the two arms. Two such screens are parallel measurements of the same condition, so their correlation over genes estimates the reliability of one served value directly, and the ceiling is its square root. The median reliability is 0.611 and the median ceiling 0.781, with the heterozygous arm at 0.797 and the homozygous arm at 0.758. Against that ceiling the best baseline of Table 10 reaches 24 percent.

This estimate is not the same quantity as the served-error ceilings elsewhere in the document, and the difference runs in the conservative direction. A cross-screen repeat carries the batch variation that two wells in one plate do not, so it measures reliability against everything that changes between screens. That is closer to what a model trained across screens faces, and closer to Vanacloig’s three-batch replicate agreement than to a within-plate standard error. Two cautions on it: the fourteen repeated compounds are the ones a screening campaign repeats, which are reference compounds rather than a random sample of the panel, and the pairs are spread unevenly, with methotrexate alone contributing 30 of the 71.

Table 9: Recovering a ceiling for Hoepfner. Experiments deposited and repeated columns come from a census of every column header in the two score matrices: no experiment appears twice, so the technical replicates are not separately deposited. The condition pairs are the same compound at the same concentration in two different studies; for parallel measurements the observed correlation estimates the reliability of one value and the ceiling is its square root. The last column is the best baseline of Table 10 as a fraction of that ceiling.

arm	experiments deposited	repeated columns	condition pairs	compounds	reliability	ceiling	of ceiling
HIP	2,905	0	35	13	0.636	0.797	0.24
HOP	2,904	0	36	14	0.574	0.758	0.25
both	5,809	0	71	14	0.611	0.781	0.24

Table 10: Compound cold-start baselines per dataset on the centered target, medians over folds. The null is the similarity-weighted mean of randomly chosen training compounds. LOCO is leave-one-compound-out; the alternative is ten folds grouped by compound, and the two are not directly comparable. A dataset with no served uncertainty has no ceiling and its score is reported alone. Eleven of the twelve encoders are scored on the four partner datasets, because the three-dimensional encoder cannot embed every compound in them.

dataset	scheme	folds	null	best ridge	best kNN	best overall	model, encoder	ceiling	of ceiling
Vanacloig 2022	LOCO	41	0.028	0.311	0.284	0.311	ridge, fcfp4 count	0.84	0.37
Hillenmeyer HOM	10fold	10	0.030	0.105	0.141	0.141	knn5, fcfp4 count	0.94	0.15
Hillenmeyer HET	10fold	10	0.003	0.113	0.126	0.126	knn1, maccs	0.79	0.16
Hoepfner 2014	10fold	10	0.023	0.189	0.169	0.189	ridge, fcfp4 count	none served	
Wildenhain 2015	10fold	10	0.168	0.249	0.394	0.394	knn5, ecfp4 count	0.87	0.45

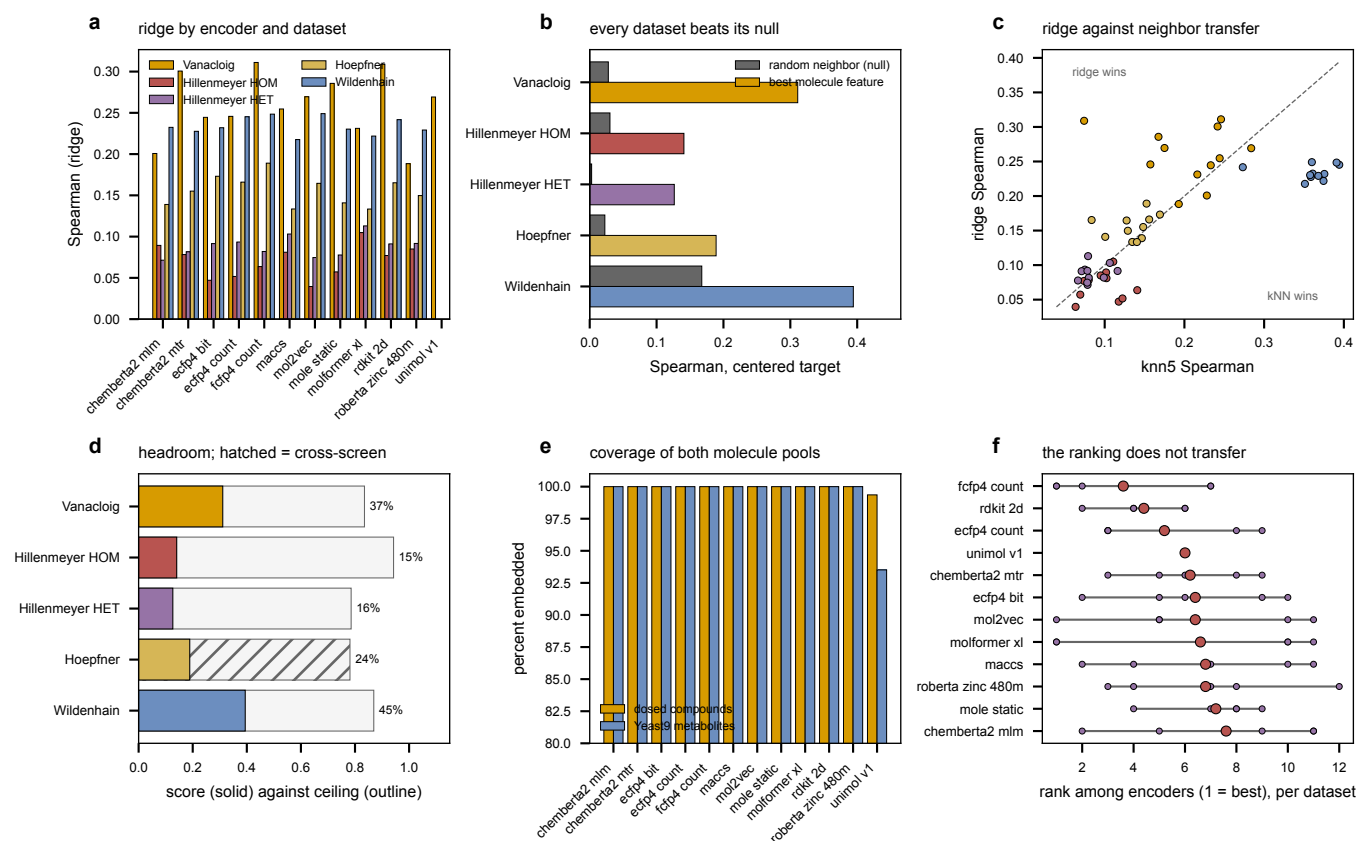


Figure 5: Molecule encoders on the compound cold-start, centered target. **(a)** Every encoder and dataset under ridge. **(b)** The best molecule feature against the no-feature null per dataset; every dataset clears its null, and Wildenhain’s null is by far the highest because a random neighbor among 5,170 compounds is already informative. **(c)** Ridge against nearest-neighbor transfer, one point per dataset and encoder; points below the diagonal are datasets where neighbor transfer wins, which is every Wildenhain point and most Hillenmeyer points. **(d)** The best score against the ceiling. The Hoepfner bar is hatched because its ceiling comes from the cross-screen estimator of Table 9 rather than from a served uncertainty, so it is not the same quantity as the other four. **(e)** What share of each molecule pool an encoder can embed: the 5,472 dosed compounds and the 894 structure-bearing Yeast9 metabolites. Only the three-dimensional encoder falls short, on the ions and metal salts it cannot build a conformer for, so coverage is not what separates these encoders. **(f)** Each encoder’s rank among the twelve on each dataset, the line spanning its best and worst rank and the large mark its mean. The spans overlap almost completely, so the encoder that wins on one dataset is not the encoder that wins on another and no single choice is supported across all five.

6 The genotype axis resolves ■

A homozygous diploid deletion can be written as the haploid genotype plus a statement of ploidy, because both copies are gone. A heterozygous deletion cannot, because one copy survives and the gene sits at half dose. The question is whether the schema can express that and whether the modeling stack can consume it. The two answers differ.

The schema already carries it. Heterozygosity is stored in two parts. A genome-wide baseline sits on the reference genome’s ploidy field, which is the copy number of every unperturbed gene, and a per-locus deviation sits on an engineered copy-number perturbation carrying the copies present and the copies in the reference. That extension was made for exactly this case. Read from the built stores, all five datasets round-trip: Vanacloig and Wildenhain are haploid with deletion leaves, the Hillenmeyer homozygous arm and the Hoepfner homozygous arm are diploid with deletion leaves, and the Hillenmeyer heterozygous arm and the Hoepfner heterozygous arm are diploid with a one-of-two copy-number leaf marked present. Heterozygous and homozygous records are distinguishable from the stored record alone on three redundant signals: the perturbation class, the state field, and whether the copy-number fields exist.

The tensor layer discards it, and that is a correctness problem. The processor that turns a genotype into model input reads exactly one field off every perturbation, the systematic gene name. Neither copy number nor ploidy appears anywhere in the dataset-to-tensor layer. A heterozygous record and a homozygous record for the same gene therefore produce byte-identical input, one index and one boolean each.

That matters quantitatively because Hoepfner measures both. Its two arms share 4,885 genes and 147 compounds, and **648,977 gene-by-compound cells are measured both heterozygous and homozygous**. Under the current representation those 648,977 pairs are identical inputs carrying two different measured responses, which is label noise rather than a missed feature.

One channel closes it. A per-gene functional dose, copies present over copies in the reference, is lossless for these five datasets on the dosage axis, because every perturbation in all five is either a full deletion or a one-of-two engineered variant. It needs no separate ploidy feature, because the vector’s background level is the ploidy: a haploid deletion sits at zero against a background of one and a homozygous diploid deletion sits at zero against a background of two, so the two remain distinguishable. It is a strict generalization of the present boolean, since every current dataset emits zero for a deletion and reproduces current behavior exactly.

Three things the channel does not carry, all measured rather than supposed. Cassette identity is lost, and this is not a detail: a KanMX replacement inserts a heterologous gene that is not in the S288C universe, so no vector over yeast genes can hold it. Barcode and collection are lost, and they are the readout identity in a pooled assay. And perturbation cardinality becomes a strain property rather than a constant, since Vanacloig contributes four indices per record against one for the others. Allele classes off the dosage axis, temperature-sensitive, DAmP and CRISPR interference, sit at full copy number and are functionally compromised, so a dose vector cannot hold them either; none of the four datasets contains one, and they are out of scope for this phase (Section 1). When they are needed the natural extension is a second per-gene channel for the expression-modulation axis the schema already distinguishes from the dosage axis.

A flagged risk for the metabolic module. The flux layer hard-zeros a deleted gene’s slot in the genome-scale model. A heterozygous deletion is not a zero there, so admitting heterozygous data without changing that step asserts a full knockout of a gene that still has a working copy. The fix is the same quantity the operator uses: the layer reads the functional dose d_i rather than a presence bit (Section 11).

7 The dose axis does not resolve ■

A pooled model sees one dose field, and a field that means different things in different rows is a nuisance variable rather than a feature. The five datasets use three mutually non-convertible regimes, and the measurement is worse than a missing column.

Vanacloig’s dose is a potency, not a quantity. Of its 143,218 records, 3,492 carry a numeric concentration and they are all one compound, benomyl. The other 40 of 41 compounds carry a target-inhibition basis with the concentration unstated. There is no molar value to pool.

Wildenhain’s dose is a constant. Every one of its 5,170 compounds sits at the same fixed concentration, so its dose column has zero variance and carries no information at all. This is harmless for a model that ignores dose and fatal for one that regresses on it.

Hillenmeyer is the only arm with real variation. Its two arms carry a numeric dose on essentially every record, converting to molar for 84 and 96 percent of them, across a range spanning nine orders of magnitude, with up to seven doses per compound.

Hoepfner is the only possible bridge, and it is two compounds wide. It is the only dataset stating both a molar concentration and a target-inhibition basis, on 100 percent of its records, with a median of two doses per compound. In principle that calibrates what the basis means in molar terms, which is exactly what Vanacloig lacks. In practice Vanacloig and Hoepfner share two compounds, so the calibration cannot reach the panel.

Shared compounds are not dosed alike. Across the five datasets, 214 compounds are dosed in more than one with a molar value. The median ratio between the highest and the lowest of those doses is 12.5, and the 90th percentile is 100. A compound shared by two datasets is typically dosed an order of magnitude apart in them.

What follows. Dose cannot be one shared continuous feature. Three routes exist and they are not equivalent. A dose-basis token per dataset alongside log molar where it exists keeps the information and accepts that the model cannot compare across bases. Dropping absolute dose and training on within-condition gene profiles removes the problem and makes Wildenhain’s constant harmless. Learning a per-compound potency scale from the multi-dose compounds and using it to place the single-dose ones is a modeling project rather than a preprocessing step. The representation in Section 12 takes the first, because it is the only one that keeps the information without a new model.

8 What pooling buys chemically ■

Pooling is worth the trouble only if the partner panels put chemistry near the application panel. Table 11 measures that for every Vanacloig compound: its most similar compound in each partner, under Tanimoto on count ECFP4. The fingerprint metric is the one reported because cosine between dense embeddings is high for any two molecules and says nothing about closeness.

Table 11: Coverage of the 41-compound Vanacloig panel by each partner. Exact counts identical InChIKeys; the median nearest neighbor and the threshold counts use non-exact neighbors only, so an exact match cannot inflate them.

source	exact	median NN	above 0.5	above 0.7
union of others	12	0.600	13	3
Wildenhain 2015	10	0.545	13	3
Hillenmeyer HOM	2	0.327	1	0
Hillenmeyer HET	2	0.317	9	1
Hoepfner 2014	2	0.305	0	0

Wildenhain supplies essentially all of the gain. Exact overlap with the panel rises from two compounds against either Hillenmeyer arm to ten against Wildenhain and twelve against the union, and the median nearest neighbor rises from 0.33 and 0.32 to 0.60. Hoepfner contributes nothing chemically: two exact matches, a median nearest neighbor of 0.29, and no non-exact neighbor above 0.5.

The gain is uneven, and it fails where the application needs it. Sorting the hydrolysate panel by its nearest neighbor in the union of the other four, ferulic acid becomes an exact match, vanillin reaches 0.62 against it, syringaldehyde 0.60 against syringic acid, and coumaric acid 0.55 against caffeic acid. The phenolics are genuinely covered. Isobutanol improves from 0.14 against sorbitol, which was its best neighbor when only Hillenmeyer was available, to 0.47, but its neighbor is an amino alcohol rather than an alcohol. Furfural sits at 0.30 against benzaldehyde and ethanol at 0.31. At two and five heavy atoms the small alcohols are nearly empty under a substructure fingerprint, so this is a limit of the representation as much as of the panel.

Pooling therefore buys phenolic coverage and does not buy alcohol coverage, and isobutanol is the application target.

9 Molecular encoders and their coverage ■

A compound enters the model as a vector, and the choice of encoder is a modeling decision made on measured coverage and measured behavior rather than on a reading list. Twelve small-molecule encoders live in `torchcell/molecule/`: a `MoleculeEncoder` turns a list of SMILES into one fixed-width float32 vector per molecule, raises on a SMILES it cannot parse rather than returning zeros, and is deterministic across calls. Four are RDKit fingerprints and descriptors with no learned weights (count and bit ECFP4 at 2,048, feature-class FCFP4, MACCS keys, and the 217 RDKit 2D descriptors); one is Mol2Vec, a skip-gram over Morgan substructure identifiers; four are SMILES transformers (the two ChemBERTa-2 heads, MolFormer-XL, a ZINC RoBERTa); one is Uni-Mol, which embeds an ETKDG conformer; and one is MolE, the contrastively pretrained graph network built for antimicrobial activity. Weights that are not on the Hugging Face hub (Mol2Vec, MolE) sit under `\$DATA\_ROOT` with a pydantic manifest recording the source URL, the retrieval command and the sha256, re-hashed on every load. The registry and its tests are described in the dendron note `torchcell.molecule.encoders`.

Why these twelve. Three principles chose the set. The benchmark of 25 pretrained embeddings on 25 datasets found almost every one statistically indistinguishable from count ECFP [▲](#) [external: arXiv 2508.06199, September 2026 revision], so fingerprints are the mandatory baseline and every learned encoder has to earn its place against them. Everything installs from RDKit, pip or the Hugging Face hub, with one vendored exception. And the four representation families are each present, so a comparison can say which family carries chemistry-to-response signal rather than which checkpoint won (Table 12).

- **Fingerprints and descriptors** (count and bit ECFP4, FCFP4, MACCS, the 217 RDKit descriptors): no training corpus, hashed or curated substructures and physicochemical properties [▲](#) [external: Rogers and Hahn 2010, 10.1021/ci100050t; Durant et al. 2002, 10.1021/ci010132r]. FCFP4 tests whether pharmacophore feature classes beat atom identity; the descriptors are what a tolerance biologist reasons with (logP, molecular weight, polar surface area).
- **Mol2Vec**: skip-gram over Morgan identifiers on 19.9 million ZINC and ChEMBL compounds, 300 dimensions, still the metabolomics workhorse [▲](#) [external: Jaeger, Fulle and Turk 2018, 10.1021/acs.jcim.7b00616].
- **SMILES transformers**: ChemBERTa-2 in its masked-language and multitask-regression heads on 77 million PubChem SMILES [▲](#) [external: Ahmad et al. 2022, arXiv 2209.01712], MoLFormer-XL on 1.1 billion PubChem and ZINC SMILES with linear attention, of which the public checkpoint carries the 10 percent subset [▲](#) [external: Ross et al. 2022, 10.1038/s42256-022-00580-7], and a community ZINC RoBERTa at 480 million SMILES added for availability. The two ChemBERTa heads share a backbone, which separates SMILES syntax from property supervision.
- **3D**: Uni-Mol, pretrained on 209 million conformers by atom masking, coordinate denoising and pair-distance prediction [▲](#) [external: Zhou et al. 2023, ICLR]; the one encoder that can say whether conformation matters for the small solvents and the phenolics.
- **Graph**: MolE, a GIN pretrained with a Barlow Twins objective on 100,000 PubChem structures and validated experimentally for antimicrobial discovery [▲](#) [external: Olayo-Alarcon et al. 2025, 10.1038/s41467-025-58804-4]; whole-cell growth inhibition is the published task closest to inhibitor tolerance, and its model code is vendored under `third\party\mole/`.

Table 12: Provenance of the twelve encoders: representation family, training corpus, objective and source. [▲](#) Temporary: the linked works are not yet in the library mirror and are given by their public link; the flag lifts when they are mirrored and cited through the bibliography.

encoder	family	trained on	objective	source
ecfp4_count, ecfp4_bit	fingerprint	none; hashed radius-2 atom environments	none	Rogers and Hahn 2010
fcfp4_count	fingerprint	none; pharmacophore feature invariants	none	Rogers and Hahn 2010
maccs	fingerprint	none; 166 curated substructure keys	none	Durant et al. 2002
rdkit_2d	descriptors	none; 217 physicochemical descriptors	none	RDKit 2026.03
mol2vec	fragment embedding	19.9M ZINC and ChEMBL compounds	skip-gram over Morgan identifiers	Jaeger, Fulle and Turk 2018
chemberta2_mlm	SMILES transformer	77M PubChem SMILES	masked language modeling	Ahmad et al. 2022
chemberta2_mtr	SMILES transformer	77M PubChem SMILES	multitask regression on RDKit properties	Ahmad et al. 2022
molformer_xl	SMILES transformer	1.1B PubChem and ZINC SMILES; public checkpoint is the 10 percent subset	masked language modeling, linear attention	Ross et al. 2022
roberta_zinc_480m	SMILES transformer	480M ZINC SMILES	masked language modeling	Hugging Face entropy-/roberta_zinc_480m
unimol_v1	3D	209M conformers	atom masking, coordinate denoising, pair distances	Zhou et al. 2023 (ICLR)
mole_static	graph (GIN)	100k PubChem structures	Barlow Twins on masked sub-graph views; validated for antimicrobial activity	Olayo-Alarcon et al. 2025

Left out. Chemprop’s D-MPNN is trained end to end and belongs to the model phase as an architecture, not to an embedding comparison. CLAMP, the strongest frozen model in the benchmark, aligns fingerprints to mammalian and enzymatic assay text and ships only as a research repository; it is the first candidate for a second pass if the fingerprint arm wins. MolCLR, GraphMVP, 3D Infomax, Mole-BERT and KPGT are GIN checkpoints of MolE’s family in research repositories; GEM is PaddlePaddle only; CLOOME, MoCoP and InfoAlign encode human Cell Painting or expression phenotypes whose transfer to yeast growth is untested; MoleculeSTM, MolBERT, R-MAT and CDDD are covered by the

SMILES family already present. Reaction-aware encoders were left out on purpose: a reaction-hypergraph embedding places a metabolite by the reactions it takes part in, which is the right context for a Yeast9 metabolite and no context at all for an inhibitor whose fate in the cell is not in the model. The metabolites among the medium components can carry a reaction branch (Section 10); the inhibitors cannot.

Two caveats for these molecules. The corpora are drug-like, 20 to 40 heavy atoms. Ethanol and isobutanol have 3 and 5, so ECFP is nearly empty and a transformer embedding is poorly determined for them, which is part of why isobutanol has no close chemical neighbor in any partner (Section 8). The salts (sodium acetate, the imidazolium chlorides, the metal chlorides) are multi-fragment SMILES that fingerprints handle and Uni-Mol cannot conformer.

Coverage over the five datasets. Table 13 gives every encoder over the dosed compounds of all five datasets, 5,472 distinct InChIKeys in the union. Every one of them now has a structure, and that sentence hides a fix worth recording. The curated identity table has a row for every Vanacloig, Hillenmeyer and Wildenhain compound, but Hoepfner derives 88 of its 148 InChIKeys from the SMILES released in its Table S1 and those keys were never curated, so until the served record’s own SMILES was read as the structure source those 88 compounds, 60 percent of Hoepfner’s chemistry, were invisible to every encoder. The precedence is the curated table where it has a row and the served record where it has none, because the two routes measurably disagree on the stereo block for some compounds. Eleven of the twelve encoders embed the whole union. Uni-Mol refuses the compounds for which no three-dimensional conformer exists, the isolated ions and metal salts, each named in `results/embeddings/failures.json`; its wrapper library would otherwise substitute flat or all-zero coordinates and serve a vector from nothing, which is why the encoder’s `check` reruns the conformer step and raises. One dosed Hillenmeyer name, tunicamycin, is a homologue mixture with no InChIKey and is not a compound here.

Table 13: Coverage of every encoder over the dosed compounds of the five datasets (distinct InChIKeys with a dosed record). The last column is wall time in seconds to embed the 5,472 compounds of the union on one shared GPU, model load excluded.

encoder	dim	Vanacloig	HOM	HET	Hoepfner	Wildenhain	s
chemberta2_mlm	384	41 of 41	114 of 114	290 of 290	148 of 148	5170 of 5170	3.4
chemberta2_mtr	384	41 of 41	114 of 114	290 of 290	148 of 148	5170 of 5170	3.4
ecfp4_bit	2,048	41 of 41	114 of 114	290 of 290	148 of 148	5170 of 5170	1.3
ecfp4_count	2,048	41 of 41	114 of 114	290 of 290	148 of 148	5170 of 5170	1.4
fcfp4_count	2,048	41 of 41	114 of 114	290 of 290	148 of 148	5170 of 5170	1.8
maccs	167	41 of 41	114 of 114	290 of 290	148 of 148	5170 of 5170	4.9
mol2vec	300	41 of 41	114 of 114	290 of 290	148 of 148	5170 of 5170	1.9
mole_static	1,000	41 of 41	114 of 114	290 of 290	148 of 148	5170 of 5170	4.9
molformer_xl	768	41 of 41	114 of 114	290 of 290	148 of 148	5170 of 5170	8.0
rdkit_2d	217	41 of 41	114 of 114	290 of 290	148 of 148	5170 of 5170	41.5
roberta_zinc_480m	768	41 of 41	114 of 114	290 of 290	148 of 148	5170 of 5170	9.5
unimol_v1	512	41 of 41	103 of 114	283 of 290	136 of 148	5163 of 5170	807.3

Cost. The fingerprints, Mol2Vec, MolE and the SMILES transformers embed the union in seconds; the RDKit descriptors take longer only because of a few slow descriptors; Uni-Mol takes minutes because of conformer generation. At the scale of these datasets no encoder is expensive, and the choice is made on what the vectors do. The same twelve encoders also cover the 37 medium components and the 894 structure-bearing Yeast9 species (Section 10), so one embedding table serves every molecular slot in the representation.

Chemical similarity against response similarity. Table 14 measures, for every encoder and both Hillenmeyer arms, whether two conditions that are chemically similar respond similarly across datasets: the Spearman between chemical similarity (Tanimoto for fingerprints, cosine for dense embeddings) and the cross-dataset response Spearman over every Vanacloig-by-partner condition pair. Against the homozygous arm the correlation is 0.08 to 0.09 for count and bit ECFP4, MolFormer-XL and Mol2Vec, 0.05 to 0.06 for the ChemBERTa-2 heads, and absent for MolE, the RDKit descriptors and the ZINC RoBERTa; against the heterozygous arm every value is below 0.04. The effect is real and small, because almost every pair sits at a Tanimoto below 0.25 where response similarity is centered on zero. The cosine medians of the dense encoders, 0.8 to 0.95, are not comparable to Tanimoto: cosine between mean-pooled transformer states is high for any two molecules, so only the fingerprint rows say how close the chemistry is.

Read together with Section 4, the encoders divide by where they are asked to work. Within Vanacloig, where medium, dose basis and readout are constant, a ridge map from a fingerprint reaches 0.31 on the compound-specific target. Across datasets the same fingerprints reach 0.09. No encoder in the set closes that gap, which is the measured reason the representation carries the dataset and dose-basis tokens rather than asking chemistry to bridge regimes on its own.

Table 14: Per encoder and Hillenmeyer arm: the median similarity of each Vanacloig compound to its nearest partner compound, the Spearman between chemical similarity and cross-dataset response Spearman over every condition pair with its p value, and the median response Spearman of the top decile of pairs by chemistry against the rest.

encoder	partner	NN median	$\rho(\text{chem, response})$	p	top-decile ρ	rest ρ
chemberta2_mlm	HET	0.85	0.037	4.8e-05	0.005	0.002
chemberta2_mtr	HET	0.94	0.031	6.7e-04	0.005	0.002
ecfp4_bit	HET	0.33	0.014	1.4e-01	0.003	0.002
ecfp4_count	HET	0.32	0.033	3.3e-04	0.004	0.002
fcfp4_count	HET	0.44	0.039	2.6e-05	0.005	0.002
maccs	HET	0.60	0.002	8.6e-01	0.003	0.002
mol2vec	HET	0.94	0.015	9.8e-02	0.003	0.002
mole_static	HET	0.93	-0.005	5.6e-01	0.002	0.002
molformer_xl	HET	0.82	0.040	1.1e-05	0.006	0.002
rdkit_2d	HET	0.72	0.036	7.2e-05	0.005	0.002
roberta_zinc_480m	HET	0.85	0.011	2.1e-01	0.003	0.002
unimol_v1	HET	0.90	0.033	3.4e-04	0.004	0.002
chemberta2_mlm	HOM	0.85	0.064	1.3e-05	0.019	0.011
chemberta2_mtr	HOM	0.93	0.054	2.4e-04	0.015	0.011
ecfp4_bit	HOM	0.33	0.089	1.3e-09	0.011	0.012
ecfp4_count	HOM	0.33	0.080	4.7e-08	0.014	0.011
fcfp4_count	HOM	0.39	0.038	9.8e-03	0.010	0.012
maccs	HOM	0.55	0.041	5.4e-03	0.014	0.011
mol2vec	HOM	0.95	0.078	8.8e-08	0.020	0.011
mole_static	HOM	0.91	0.008	5.7e-01	0.011	0.012
molformer_xl	HOM	0.81	0.079	7.0e-08	0.022	0.011
rdkit_2d	HOM	0.71	0.016	2.7e-01	0.012	0.011
roberta_zinc_480m	HOM	0.83	-0.028	5.6e-02	0.009	0.012
unimol_v1	HOM	0.90	0.042	6.4e-03	0.012	0.011

10 Whether the metabolites can join ■

The plan is to represent media components and metabolic-model metabolites as molecules in the same embedding space as the dosed inhibitors. Two things decide whether that works: how many Yeast9 species have a structure at all, and whether the encoders can embed them.

The encoders are not the constraint. Eleven of the twelve registered encoders embed every structure-bearing species with no failures (Table 15). The three-dimensional encoder reaches 836 of 894, its refusals being monatomic ions with no conformer and planar or linear molecules, together with the inositol-phosphoceramide sphingolipids. The descriptor block emits undefined entries on 18 rows, so a caller imputes before taking any distance.

Table 15: Encoder coverage over the 894 structure-bearing Yeast9 species.

encoder	dim	n_attempted	n_embedded	n_failed
ecfp4_count	2048	894	894	0
ecfp4_bit	2048	894	894	0
fcfp4_count	2048	894	894	0
maccs	167	894	894	0
rdkit_2d	217	894	894	0
mol2vec	300	894	894	0
chemberta2_mlm	384	894	894	0
chemberta2_mtr	384	894	894	0
molformer_xl	768	894	894	0
roberta_zinc_480m	768	894	894	0
unimol_v1	512	894	836	58
mole_static	1000	894	894	0

The structures are the constraint. Yeast9 holds 2,806 compartment-specific entries that collapse to 1,378 distinct species, and only 894 of those carry a SMILES, all from the release’s own name-keyed table. No species carries an InChI, so every key used here is derived from the shipped SMILES rather than sourced. Two quality problems matter for a shared space. Every shipped SMILES is stereochemically flat, with 621 of 894 carrying an unassigned stereocenter, so L and D forms collapse onto one key. And 36 do not encode the species the model declares, judged against the model’s own formula, two of them being outright wrong molecules.

The 484 species without a structure split into two halves that need different treatment (Table 16). About 167 have no single structure and never will, being transfer RNAs, pooled lipid classes, protein species, glycan accessions and biomass pseudo-metabolites. Those need a non-molecular slot regardless of curation. The remaining 317 have a definite

structure absent from the release table, dominated by acyl-resolved lipids, and 295 of the 484 carry a cross-reference identifier, so a lookup route exists.

Table 16: Yeast9 species by structure class, with how many carry a SMILES.

structure_class	n_species	n_compartment_copies	n_with_smiles
specific_molecule	914	1742	858
acyl_resolved_lipid	258	692	0
trna	44	75	0
inorganic_or_ion	31	125	28
generic_r_group	29	42	5
protein_species	24	33	2
pooled_lipid_class	23	41	0
kegg_glycan	22	22	0
acp_thioester	16	16	0
pseudo_metabolite	11	12	0
no_formula_other	6	6	1

The consequence for anything reaction-level is that 2,156 of 4,131 reactions, 52 percent, have a structure for every participant.

The axes are related but not the same. Against the full five-dataset compound set, 49 exact keys and 103 connectivity skeletons are shared with Yeast9, so 103 metabolites are dosed as exogenous compounds somewhere in the five datasets. They are the central cofactors and metabolites rather than the drug-like library. The media axis is a different story: 30 of 37 media components are Yeast9 metabolites, which is expected since media components are metabolites by construction and is the strongest argument for one shared space. Because Yeast9 is stereo-flat while the curated compound keys carry stereochemistry, any join between the axes is on the connectivity skeleton with the ambiguity recorded.

11 Where the channels enter the cell graph transformer ■

The representation has to land in a model that already exists, and the model’s own structure decides where each channel can go. This section states what the cell graph transformer does today, read from `torchcell/models/equivariant\_cell\_graph\_transformer.py`, and then places the genotype dose, the environment context and the dataset token in it. Anything not yet run is labeled as a hypothesis.

The stack as it stands. The encoder holds one token per gene, $N = 6,607$, plus a CLS token, $H^{(0)} \in \mathbb{R}^{(N+1) \times d_h}$, from a learnable table or from precomputed sequence embeddings. L transformer layers follow, and in each layer every attention head is pulled toward one of the nine gene-gene graphs by a divergence penalty $\text{KL}(A_g \parallel \alpha^{(\ell,k)})$ of weight λ , or restricted to that graph’s edges by a hard mask. The perturbation then enters through a Type I operator, representation to representation: the perturbed genes’ current embeddings are the keys and values of a cross-attention over every gene token,

$$H_{\text{pert},i} = h_i + \sum_{j \in S} \alpha_{ij} W_V h_j,$$

so the output keeps its $[\text{batch}, N, d_h]$ shape and is equivariant over genes. The operator is neither a mask nor a divergence term. It is soft by construction, and every perturbation in it is a set membership, $j \in S$: the model knows which genes are perturbed and nothing about how much. Type II readouts then map H_{pert} to outputs, and the existing ones are the instruments the earlier experiments built: a pooled perturbation head for gene interaction (experiment 010), a global head for fitness, a per-gene head for expression and proteome that keeps the per-node structure and can emit distributional parameters (019), and a per-metabolite head that pools gene embeddings over the gene-reaction-metabolite incidence of Yeast9 (026 and 027), with a masked multitask loss that scores each head only on the records carrying its phenotype. Experiment 030 added a per-entry dataset token concatenated to the head input at the readout.

Genotype: the dose scales the operator. The operator’s set membership generalizes to a scaled membership without changing its form. With $m_j = 1 - d_j$ the perturbation magnitude at gene j , one for a deletion and one half for a heterozygous deletion in a diploid,

$$H_{\text{pert},i} = h_i + \sum_{j \in S} \gamma(m_j) \alpha_{ij} W_V h_j, \quad \gamma(1) = 1.$$

The constraint $\gamma(1) = 1$ is what makes this the smaller of the two changes: every record in every current dataset is a deletion with $m_j = 1$, so every trained model and every earlier result is reproduced exactly, and the only new behavior is at $m_j = \frac{1}{2}$. Whether γ is a fixed identity, $\gamma(m) = m$, or a learned scalar map is an arm, not a design choice; the

648,977 Hoepfner cells measured in both arms are the data that decide it, and they are the only such data. *Hypothesis (untested)*: a heterozygote attenuates the deletion response rather than producing a different one, which is what a scalar γ can express and a separate feature would have to learn. The same d_i replaces the presence bit the flux layer uses to zero a deleted gene’s reactions, so the metabolic branch stops asserting a full knockout of a gene with a working copy.

Environment: nothing reaches the tensor layer today. The dataset-to-tensor processor reads exactly one field off every perturbation, the systematic gene name, and no environment field at all: no compound, no medium, no dose appears anywhere in `torchcell/data/graph\_processor.py`. Every chemogenomic record therefore looks to the current model like a deletion with no condition. The environment context z_E of Section 12, the pooled compound embeddings, the pooled medium embeddings, the dose-basis one-hot and the masked log molar value, has two possible entry points and they are the first two arms.

The first is the readout: z_E is concatenated to the Type II head input beside the dataset token, exactly as 030 concatenates the token. This is the smallest change and it leaves the encoder and the operator untouched, so the gene representation is condition-free and the head does all the conditioning. The second is the operator: the compound embeddings are projected to d_h and appended as extra key-value rows of the cross-attention, so every gene token attends to the compound as it attends to the perturbed genes, and the compound shapes the perturbed representation itself. *Hypothesis (untested)*: the second arm is the one that can represent a gene-by-compound interaction, which the readout arm can only approximate through the pooled z_S , and it is where the 0.31-against-0.09 gap of Section 4 says the signal lives; the first arm is the control that says how much the readout alone recovers.

One decoder over four response scales. The four kept datasets report a response on three scales, a log ratio, a z score and a sensitivity score, from two assays (Table 2). Three decoder designs are available. Four heads, one per dataset, under the existing masked multitask loss: this is what the code does today for different phenotype types, and it shares nothing at the output. One head with a per-dataset output calibration, a learned affine map per source: it shares the head and lets each source have its scale. Or one head with the dataset token at the readout, the 030 mechanism: the token is projected and concatenated to the head input, so the head learns each source’s scale and offset from the same weights.

The third is the one taken here, for a measured reason and a structural one. The measured reason is the 030 smoke test on GilaHyper job 2867 (W&B run 5xn8w427): a synthetic per-source offset of 0.3 injected into twin copies of four Kuzmin sources was recovered as a prediction difference of 0.285 with standard deviation 0.053, so the token does carry a source’s residual scale. The structural reason is Section 2: every physical field is constant within a dataset, so the token is also the only channel those fields need, and four heads would have to learn them four times. Each dataset’s response is standardized to zero mean and unit variance within the training split before the loss, which the 030 label-normalization transform already does per source, so the three scales meet the loss on one footing and the token explains what standardization cannot. *Hypothesis (untested)*: whether pooling under this decoder helps Vanacloig at all, against training on Vanacloig alone, is the first question the model phase answers, and the compound-cold split of Section 4 with the reliability ceiling beside every compound is how it is scored.

The application target is 2 percent of the pool, so the run has to be able to see it. Vanacloig contributes 143,218 of the 6.39 million kept records, which is 2.2 percent. Every other dataset is between three and twenty times larger, and Section 5 measured their agreement with Vanacloig at 0.013 to 0.078 on shared cells. A pooled objective therefore has both the size and the independence to let the application target drift while the total loss improves, and a single pooled validation number would not show it happening.

Three provisions follow, and they are cheap enough to build before the first run rather than after a disappointing one.

1. **Log every metric per source, per epoch.** The correlation, the loss and the record count are reported for each of the four sources separately at every validation step, in addition to the pooled number. This is what makes drift visible as it happens rather than at the end, and it is the only way to tell a pooled gain that lifts every source from one that trades the small source for the large ones.
2. **Report Vanacloig against its own ceiling, always.** The pooled metric has no ceiling because the sources have different reliabilities. The application question is what fraction of the 0.84 Vanacloig ceiling the model reaches on held-out compounds, so that number is the headline and the pooled loss is diagnostic.
3. **Keep a fine-tune stage in the plan from the start.** If the joint run does not favor the application target, the remedy tested first is a short final stage on Vanacloig alone at a reduced learning rate, starting from the pooled checkpoint. The per-source curves say whether that stage is needed and the per-source metric at the end of it says whether it worked. *Hypothesis (untested)*: the pooled run supplies a gene prior that a Vanacloig-only run cannot, and a short fine-tune recovers the compound-specific fit the pooled objective dilutes. Both halves of that are measurable against the Vanacloig-only arm of the first round.

Source reweighting in the loss is deliberately NOT part of the plan yet. It is a second lever on the same problem, and adding it before the per-source curves exist would make the first round unable to say which lever mattered.

Masking against the divergence prior. The 025 graph-regularization sweep ran the nine graphs into layer one under the divergence prior at seven weights and under the hard mask, three seeds each, on the 376,732 triples of experiment 010 (`notes-tex/025-graph-reg-sweep`). Read at a fixed epoch 29 the mask sat with no penalty, 0.421 ± 0.003 against 0.425 ± 0.000 held-out Pearson, and the prior rose with λ without a peak inside the ladder, to 0.447 ± 0.001 at $\lambda = 1$. Read at each run’s best epoch, all arms lay within 0.010 of one another, and what the prior bought at the fixed reading was mostly resistance to the decline the unpenalized model showed after epoch 16. That is the evidence behind the suspicion that the mask is imposed too hard and too early rather than being wrong: a mask that starts open and closes over training, or a prior whose λ anneals upward, has not been run. *Hypothesis (untested)*: an annealed mask reaches the prior’s fixed-epoch score. It is a cheap arm to add to the first round because it changes the schedule and nothing else. The default for this phase is the prior at $\lambda = 1$, because it is the measured best at the reading that matters for a model trained to a budget rather than to a peak.

What the other instruments contribute. The chemogenomic head is one more Type II instrument on the same H_{pert} that the expression, metabolite and gene-interaction heads read, so a model trained here can be trained jointly with those datasets under the masked loss, and the environment context is new to all of them: the 019 expression data and the 010 triples were measured in one condition each and have never had a condition channel. *Hypothesis (untested)*: joint training with the deletion-fitness and expression data supplies the gene prior that Section 4 measured as the transferable part of the chemogenomic signal. That is a second-round question, after the single-task result exists.

12 The representation ■

Figure 6 is the input representation and the decoder the preceding sections argue for, drawn as the tensors each channel is. It has two per-record input channels and a per-record token, one shared molecule encoder, one operator and one decoder, and it carries its own limits.

Genotype. A per-gene functional dose over the gene universe, $d \in \{0, \frac{1}{2}, 1\}^N$ with $d_i = c_i/r_i$, copies present over copies in the reference. Zero for a deletion, one half for a heterozygous deletion in a diploid, one for an unperturbed gene. Ploidy needs no separate feature because it is the background level. The perturbed set is $S = \{i : d_i < 1\}$ and the magnitude $m_i = 1 - d_i$ is what scales the operator. This is lossless for these datasets and is validated by Hoepfner’s 648,977 paired cells.

Environment. The set of dosed compounds as InChIKeys through the shared molecule encoder f , the medium components through the same encoder, a dose-basis one-hot over target inhibition, fixed concentration and unstated, and the log molar concentration where a record has one with a mask where it does not. The basis token is the honest encoding given Section 7: a target-inhibition dose, a fixed molar dose and an unstated dose are three different statements, and a single number would assert a comparability that does not exist. The masked log molar value keeps the within-dataset dose variation that Hillenmeyer and Hoepfner do carry without pretending it compares across them. The four parts are concatenated into one context z_E , which enters at the readout in the first arm and as extra key-value rows of the operator in the second (Section 11).

Dataset token. A one-hot over the four sources, projected and concatenated at the readout. It carries the response scale, the assay, and every physical field, since temperature, aerobicity, duration, pH and solvent are constant inside each dataset (Table 2). No separate physical channel exists for that reason; the fields stay in the record.

The encoder table. One embedding table, keyed by InChIKey, serves the dosed compounds, the medium components and the Yeast9 metabolites. It is precomputed by `embed\_compounds.py`, `embed\_media\_components.py` and `yeast9\_molecule\_coverage.py` for all twelve encoders, so the encoder is an arm selected at data-loading time and never a training-time cost. FCFP4 count is the default because it is the measured best on the compound-cold target (Table 6); the transformers and MolE are the alternatives that a first round compares against it.

What it does not carry. The dose vector drops cassette identity, barcode and collection. Dose is never one number across datasets. Alleles off the dosage axis are out of scope for this phase. And 167 Yeast9 species have no structure and need a non-molecular slot. Each of those is a measured limit rather than an anticipated one, and each bounds a specific later step: the dose limit bounds any cross-dataset dose-response claim, and the structureless species bound the metabolic branch.

Implementation, in order.

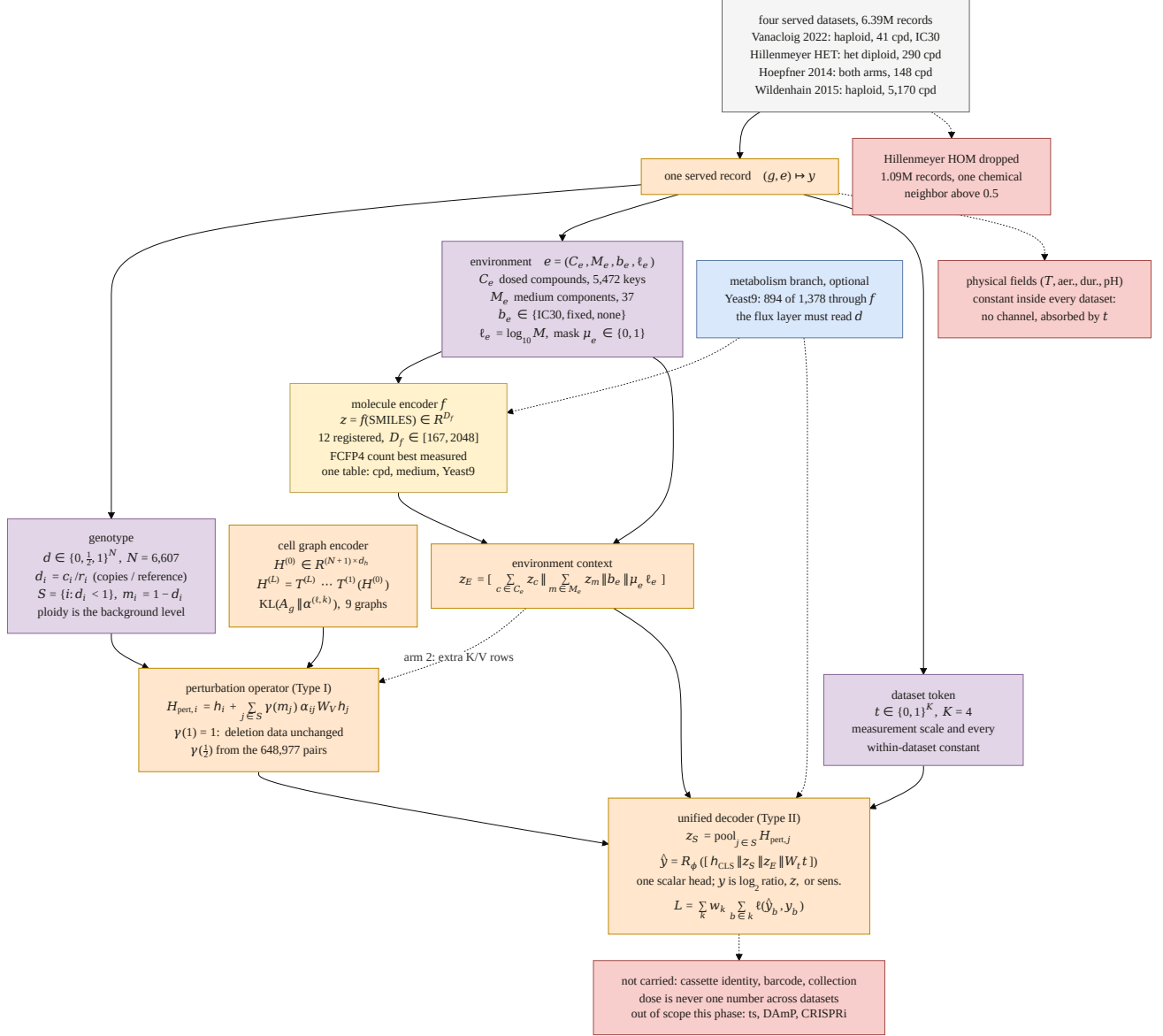


Figure 6: One input representation and one decoder for the four kept datasets, in the notation of the cell graph transformer: N genes, d_h the hidden width, S the perturbed set, H_{pert} the Type I output, R_ϕ a Type II readout. Gray is a served dataset, purple a per-record input channel, yellow the fixed embedding table, orange a computation, blue the optional metabolism branch, red what the representation drops or does not carry. The genotype is a per-gene functional dose whose background level is the ploidy, and it scales the perturbation operator through $\gamma(m_j)$. The environment is a compound set and a medium set through the shared molecule encoder, a dose-basis token and a masked log molar value, pooled into one context z_E . The dataset token t carries the measurement scale and every physical field, because each of those is constant within a dataset. The decoder is one scalar head over the concatenation.

1. The dataset-to-tensor processor emits, beside the perturbed gene indices, a per-index magnitude m_j computed from the perturbation’s copy number and reference copy number, and the reference genome’s ploidy; a deletion emits one. Every existing dataset produces the same tensors as today plus a vector of ones.
2. The perturbation operator takes the magnitude and scales each key-value row by $\gamma(m_j)$ with $\gamma(1) = 1$, first as the fixed identity $\gamma(m) = m$ and then as a learned scalar map; the flux layer reads d_i in place of its presence bit.
3. The processor emits the environment context from the served record: the compound InChIKeys and medium component keys resolved against the precomputed embedding table, the dose basis as a one-hot, and the log molar value with its mask. A record whose compound has no vector in the table raises rather than receiving zeros.
4. The processor emits the dataset index over a fixed vocabulary of the four sources, and the label transform standardizes each source’s response within the training split.
5. The head takes the concatenation of the CLS embedding, the pooled perturbed-gene summary, the environment context and the projected dataset token, as one scalar head under the masked loss.
6. The evaluation is fixed before the first run: compound-cold within Vanacloig, 41 folds, scored on the centered and the raw target with the reliability ceiling beside every compound, and a Vanacloig-only training arm as the control for pooling.

First round. Four arms on the fixed evaluation, three seeds each: Vanacloig alone; the four datasets with z_E at the readout; the four datasets with z_E in the operator; and the operator arm with the learned γ . The first two answer whether pooling helps at all, the third whether the gene-by-compound interaction needs the operator, and the fourth whether heterozygosity is an attenuation. The graph prior runs at $\lambda = 1$ throughout, and the annealed mask is a fifth arm if the round has budget for it. Every score is reported against the ceiling of its compound, and a partial run is a partial run.